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| <b>Subject Code</b>                            | AF5366                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Subject Title</b>                           | <b>Applications of Computing and Technology in Accounting and Finance II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Credit Value</b>                            | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Level</b>                                   | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Normal Duration</b>                         | One Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Pre-requisite / Co-requisite/ Exclusion</b> | AF5365 Applications of Computing and Technology in Accounting and Finance I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Role and Purposes</b>                       | <p>This subject is designed to study the new developments in accounting and finance technology. Students are provided with the knowledge and practical skills necessary to develop a strong foundation on core paradigms and algorithms of machine learning (ML), with a particular focus on applications of ML to various practical problems in Accounting and Finance. With the scientific computing skills and necessary theoretical background in AI and ML, students learn how to implement the Python/R/VBA to achieve the algorithms and apply the skills in High Frequency Modelling, Portfolio Analytics, Financial Statement Fraud, and others. Application Programming Interface (API) will be introduced for students to understand Direct Market Access (DMA) and automatic information collection, data analyses, and order execution.</p> <p>This subject contributes to the achievement of the MSc in Accounting and Finance Analytics programme learning outcomes by enabling students to apply technology and data analytics skills to solve accounting and finance problems faced in real-life situations in an ethical manner (Outcome 3).</p> |
| <b>Subject Learning Outcomes</b>               | <p>Upon successful completion of this subject, students should be able to:</p> <ol style="list-style-type: none"> <li>Understand the applications of artificial intelligence and machine learning in accounting and finance</li> <li>Implement the computing tools to achieve some AL and ML algorithms</li> <li>Understand the algorithmic trading systems and trading strategies</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Subject Synopsis/ Indicative Syllabus</b>   | <p><b>An introduction to Artificial Intelligence and Machine Learning</b></p> <p>Start from the regression function and bias variance trade-off to demonstrate their applications in accounting and finance practice.</p> <p><b>Classification/Prediction Problems</b></p> <p>Introduce the predictive models, Machine Learning, and the corresponding trading strategies with implementation using scientific computing tools.</p> <p><b>Clustering/Recommendation Systems</b></p> <p>Introduce algorithms to identify clusters in real life sample and to build recommendation systems.</p> <p><b>Algorithmic Trading Systems</b></p> <p>Utilize the Application Programming Interface (API) and existing trading strategy to systematically collect exchange data, analyze the data, and automatically execute the orders and the risk control systems.</p>                                                                                                                                                                                                                                                                                                     |

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| Teaching/Learning Methodology                                                                                                                                                                                                                                                                                       | Key concepts and techniques will be introduced through lectures. The subject places a lot of emphasis on project work. Students will be required to deliver a project which emphasizes on real-world accounting and finance issues. By completing the project, students should have hands-on experience in using the knowledge they have learnt in class to solve accounting and finance problems in practice. Students are encouraged to share their views and experiences actively with their lecturer and classmates. |             |                                                                                |   |          |  |
| Assessment Methods in Alignment with Intended Learning Outcomes                                                                                                                                                                                                                                                     | Specific assessment methods/tasks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | % weighting | Intended subject learning outcomes to be assessed (Please tick as appropriate) |   |          |  |
|                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             | a                                                                              | b | c        |  |
|                                                                                                                                                                                                                                                                                                                     | 1. Class participation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 5%          | √                                                                              | √ | √        |  |
|                                                                                                                                                                                                                                                                                                                     | 2. Project & Quiz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 25%         | √                                                                              | √ | √        |  |
|                                                                                                                                                                                                                                                                                                                     | 3. Mid-term test                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 20%         | √                                                                              | √ |          |  |
|                                                                                                                                                                                                                                                                                                                     | 4. Final examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 50%         | √                                                                              | √ | √        |  |
|                                                                                                                                                                                                                                                                                                                     | Total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 100 %       |                                                                                |   |          |  |
| <i>Note: To pass this subject, students are required to obtain Grade D or above in BOTH the Continuous Assessment and Examination components. In addition, the specific requirements on individual assessment components discussed above could be adjusted based on the pedagogical needs of subject lecturers.</i> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |                                                                                |   |          |  |
| Student Study Effort Expected                                                                                                                                                                                                                                                                                       | Class contact:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |                                                                                |   |          |  |
|                                                                                                                                                                                                                                                                                                                     | ▪ Lectures / Seminars                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |                                                                                |   | 39 Hrs.  |  |
|                                                                                                                                                                                                                                                                                                                     | Other student study effort:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |                                                                                |   |          |  |
|                                                                                                                                                                                                                                                                                                                     | ▪ Reading materials / textbook questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |                                                                                |   | 39 Hrs.  |  |
|                                                                                                                                                                                                                                                                                                                     | ▪ On average around 16 hours will be spent on the individual critique and around 20 hours for the group project discussion, presentation and written report                                                                                                                                                                                                                                                                                                                                                              |             |                                                                                |   | 36 Hrs.  |  |
|                                                                                                                                                                                                                                                                                                                     | Total student study effort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                                                                                |   | 114 Hrs. |  |
| Reading List and References                                                                                                                                                                                                                                                                                         | <b>References</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |                                                                                |   |          |  |
|                                                                                                                                                                                                                                                                                                                     | <i>The Next Chapter of Big Data in Finance</i> , Itay Goldstein, Chester S. Spatt, and Mao Ye, <i>The Review of Financial Studies</i> , 2025.<br><br><i>Big data, data analytics and artificial intelligence in accounting: An overview</i> , Sudipta Bose, Sajal Kumar Dey, and Swadip Bhattacharjee, <i>Handbook of Big Data Research Methods</i> , 2023.                                                                                                                                                              |             |                                                                                |   |          |  |